ABSTRACT

Submitted To,
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Title:

GroKit– An Online Grocery Management System

Objective:

The objective of the online grocery management system is to develop a smart and user-friendly web site that simplifies the grocery shopping process by allowing users to select recipes and automatically generate accurate shopping lists based on ingredient requirements and product availability. The system aims to provide time-based grocery suggestions for different meals such as breakfast, lunch, snacks, and dinner, while also suggesting suitable alternatives when items are out of stock. GroKit focuses on saving users' time, reducing manual effort, and improving shopping convenience by supporting ready-to-cook and instant food items, along with premium features like recipe videos and diet-friendly information, making it a complete and efficient online grocery solution.

Introduction:

In today's digital world, online grocery platforms have become essential for modern consumers. GroKit is designed as a complete online grocery management website that provides a seamless experience for users to select recipes and purchase grocery items from various categories. The system ensures smart shopping list generation based on ingredient requirements, real-time product availability, and time-based meal recommendations. It also supports alternative product suggestions, ready-to-cook and instant food items, and premium features such as recipe videos and diet-friendly information. This project aims to simplify grocery shopping by offering a convenient, efficient, and intelligent web-based grocery solution.

Modules:

Main Modules are

- **Admin Module:**
    - Login
    - Homepage/Dashboard
    - Master Entries
        - Location
            - District/Place
        - Product Masters
            - Category, Subcategory, Brand etc.

- Registrations Rules
    - Seller/User/Delivery/Informer
- Recipe/Diet Info Adding
- Premium plan adding
- Manage Seller
    - Accept/Reject (Send Mail)
    - Accepted List
    - Rejected List
- User List
- Seller List
- Delivery List
- Manage Complaints and Feedbacks.
- Generate Report

- **Delivery Module:**
    - Login
    - Homepage/Dashboard
    - Settings
        - Profile/Edit Profile/Change Password
    - Delivery Assign View
    - Status Updation

- **Seller Module:**
    - Login
    - Homepage/Dashboard
    - Settings
        - Profile/Edit Profile/Change Password
    - Product Upload (Product List)
        - Rate, Stock, Offer, Gallery etc.
    - View Order from Users
    - Delivery Assign
    - Update Delivery Status View
    - View Ratings, Comments etc.
    - Manage Complaints and Feedbacks.

- **User Module:**
    - Login
    - Homepage/Dashboard
    - Settings
        - Profile/Edit Profile/Change Password
    - Product View
    - Search Products
    - Add to Cart

- Payment
- View Order Status
- Delivery Updation View
- After receiving products
  - Ratings
  - Comments
- Post Complaints and Feedbacks.
- Recipe/Diet Info View
  - Ratings
  - Comments

- **Guest Module:**
  - Seller Registration
  - Informer Registration
  - User Registration
  - Delivery Registration

Technology Used:

- **Frontend:** HTML, CSS, JavaScript, Bootstrap
- **Backend:** Python + Django
- **Database:** SQLite
- **IDE:** VS Code, DB Browser
- **Tools & Libraries:** AJAX, jQuery, Font Awesome
- **Security:** Basic input validation and secure session handling

Company Profile

Company Name : Progressive Cybernetics Pvt. Ltd. (PCPL)

Industry : Information Technology & Software Services

Founded Year : 2000

Headquarters : Kochi, Kerala, India

Official Website : www.pcplglobal.com

Company Overview:

Progressive Cybernetics Pvt Ltd commenced its operations in the year 2000 at STPI in Kochi. A highly progressive and well-established enterprise in the realm of Information Technology, Progressive Cybernetics comprises of a team of focused, innovative and motivated engineering personnel, programmers, web developers, graphic designers and business analysts who deliver speedy and cost-effective IT solutions, IT sourcing, IT services and software development.

Mission:

Product mission:

To develop and sell innovative finest quality technology-based products or solutions to make life simpler and valuable for the society.

Economic Mission:

To operate the Company on a strong financial basis of profitable growth, increasing value for our stakeholders, expanding opportunities for development and growth for our employees and career aspirants.

Social Mission:

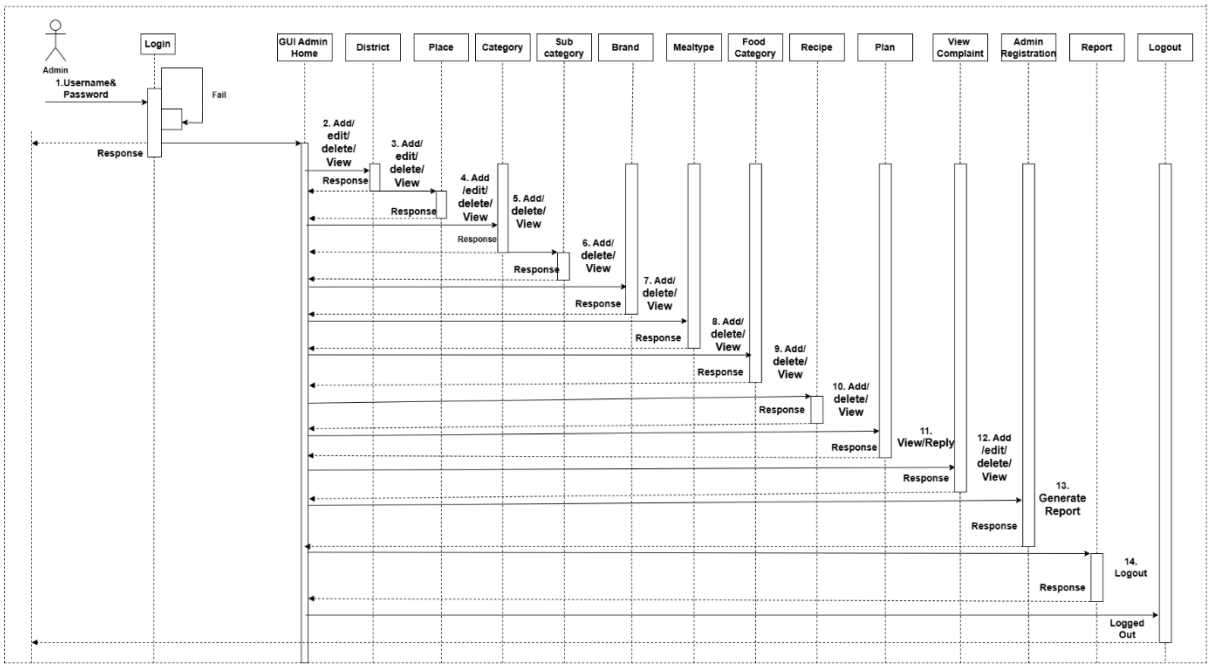
To operate the company on the belief that the pivotal role of business in society is initiating innovative ways to improve the quality of life nationally and internationally.

Vision:

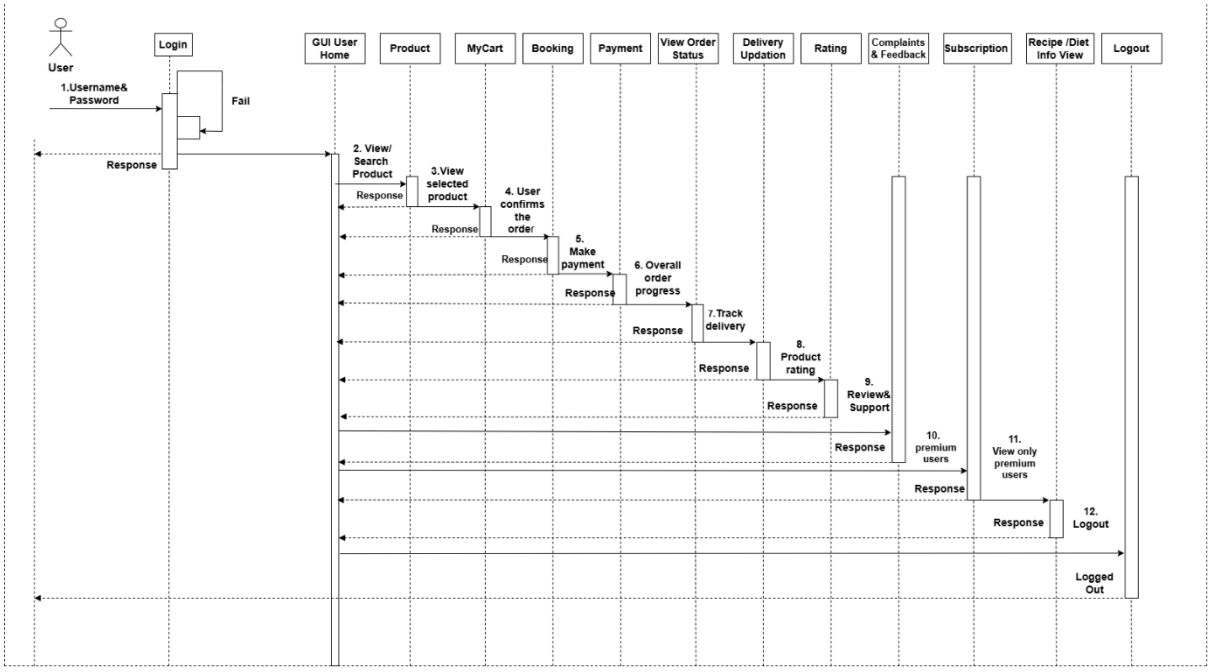
To be a partner in the progress of our country with technology-based solutions for the global world.

SSD Diagram of GroKit

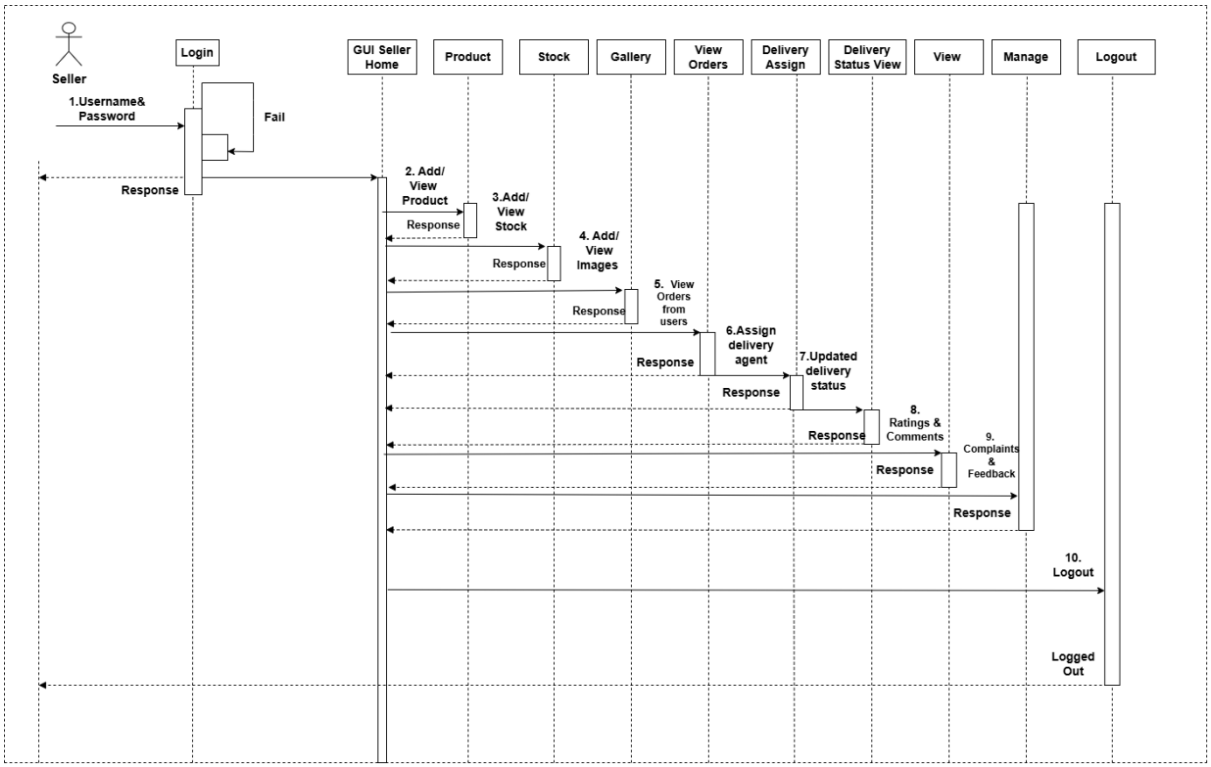
Admin- SSD



user- SSD



Seller- SSD



Delivery- SSD

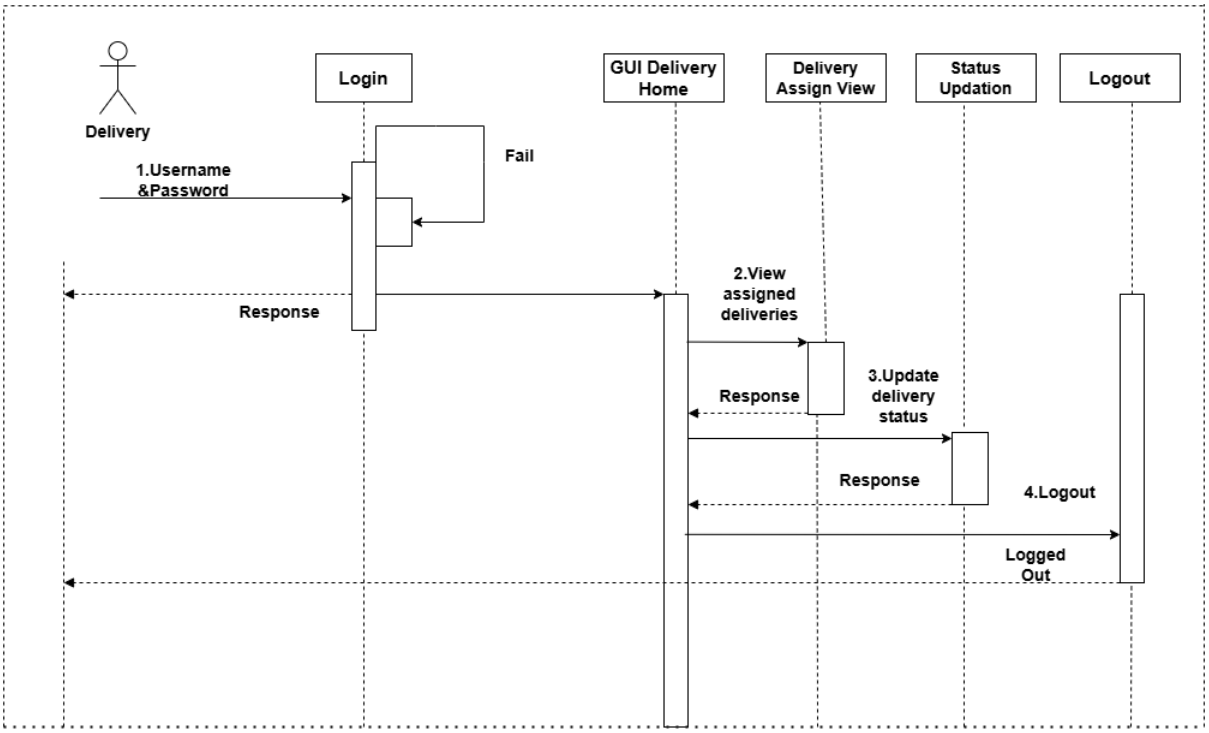


TABLE STRUCTURE

1. **Table Name: tbl_admin**
   Description: The administrator is responsible for managing users, sellers, products, and overall system operations. Each admin record is uniquely identified using admin_id.

Field Name	Data Type	Description
admin_id	Int	Stores a unique identification number for each administrator. This value is automatically generated.
name	Varchar (50)	Stores the full name of the administrator.
email	Varchar (50)	Stores the official email address of the administrator, used for communication and login.
password	Varchar (50)	Stores the encrypted password used by the administrator to log in to the system.

2. **Table Name: tbl_district**
   Description: This table stores the details of districts. District information is used to organize places, users, sellers, and delivery boys based on geographical regions.

Field Name	Data Type	Description
district_id	Int	Stores a unique identifier for each district.
name	Varchar (50)	Stores the name of the district.

3. **Table Name: tbl_place**
   Description: This table stores place information under each district. It helps in identifying the exact location of users, sellers, and delivery boys.

Field Name	Data Type	Description
place_id	Int	Stores a unique identifier for each place.
name	Varchar (50)	Stores the name of the place.

district_id	Int	Stores the district id as a foreign key to link places with districts.

4. **Table Name: tbl_category**
   Description: This table stores main product categories. Categories help to classify products for easy searching and browsing.

Field Name	Data Type	Description
category_id	Int	Stores a unique identifier for each category.
name	Varchar (50)	Stores the name of the category.

5. **Table Name: tbl_subcategory**
   Description: This table stores subcategories that belong to a particular category. It provides a finer classification of products.

Field Name	Data Type	Description
subcategory_id	Int	Stores a unique identifier for each subcategory.
subcategory _name	Varchar (50)	Stores the name of the subcategory.
category_id	Int	Stores the category id as a foreign key reference.

6. **Table Name: tbl_brand**
   Description: This table stores brand details of products. Brands help users identify the manufacturer or company of the product.

Field Name	Data Type	Description
brand_id	Int	Stores a unique identifier for each brand.
name	Varchar(50)	Stores the brand name.

7. **Table Name: tbl_gallery**
   Description: This table stores multiple images of products to provide better visual representation to users.

Field Name	Data Type	Description
gallery_id	Int	Stores a unique identifier for each gallery record.
photo	Varchar (200)	Stores the image of the product.
product_id	Int	Stores the product id associated with the image.

8. **Table Name: tbl_foodcategory**
   Description: This table is used to store different food categories. It helps in classifying recipes based on the type of food, making it easier for users to browse and search recipes.

Field Name	Data Type	Description
foodcategory_id	Int	Stores a unique identifier for each food category.
name	Varchar (50)	Stores the name of the food category.

9. **Table Name: tbl_seller**
   Description: This table stores the personal, contact, and verification details of sellers. Sellers can add and manage products after admin approval.

Field Name	Data Type	Description
seller_id	Int	Stores a unique identifier for each seller.
name	Varchar(50)	Stores the full name of the seller.
email	Varchar(100)	Stores the email address of the seller.
contact	Bigint	Stores the mobile number of the seller.
address	Varchar(100)	Stores the residential or business address of the seller.

photo	Varchar(200)	Stores the profile photo of the seller.
proof	Varchar(200)	Stores the identity proof document uploaded by the seller.
password	Varchar(50)	Stores the login password of the seller.
status	Varchar(20)	Stores the approval status of the seller account.
doj	Date	Stores the date on which the seller registered.
place_id	Int	Stores the place id to indicate seller location.

10. Table Name: tbl_user

Description: This table stores the details of users who register and use the system to purchase products.

Field Name	Data Type	Description
user_id	Int	Stores a unique identifier for each user.
name	Varchar(50)	Stores the full name of the user.
email	Varchar(100)	Stores the email address of the user.
contact	Bigint	Stores the mobile number of the user.
address	Text	Stores the delivery address of the user.
photo	Varchar(200)	Stores the profile image of the user.
password	Varchar(50)	Stores the encrypted login password.
place_id	Int	Stores the place id indicating the user's location.

11. Table Name: tbl_stock

Description: This table is used to manage the stock details of products. It helps sellers track available quantities.

Field Name	Data Type	Description
stock_id	Int	Stores a unique identifier for each stock record.
date	Date	Stores the date when stock was updated.
count	Int	Stores the number of products available in stock.
product_id	Int	Stores the product id related to this stock.

12. Table Name: tbl_plan

Description: This table stores subscription or service plan details available to users.

Field Name	Data Type	Description
plan_id	Int	Stores a unique identifier for each plan.
name	Varchar(50)	Stores the name of the plan.
duration	Varchar(30)	Stores the validity period of the plan.
details	Varchar(100)	Stores complete information about the plan features.
amount	Double	Stores the cost of the plan.

13. Table Name: tbl_cart

Description: This table stores the cart details of users before placing an order.

Field Name	Data Type	Description
cart_id	Int	Stores a unique identifier for each cart entry.
quantity	Int	Stores the quantity of the selected product.
status	Varchar(20)	Stores the cart status (active/ordered).
product_id	Int	Stores the selected product id.
booking_id	Int	Stores the booking id after order confirmation.

14. Table Name: tbl_product

Description: This table stores complete information about products uploaded by sellers. Products are displayed to users for browsing and purchase.

Field Name	Data Type	Description
product_id	Int	Stores a unique identifier for each product.
name	Varchar(100)	Stores the name of the product.
details	Text	Stores detailed description of the product.
price	Double	Stores the selling price of the product.
photo	Varchar(200)	Stores the image of the product.
subcategory_id	Int	Stores the subcategory id of the product.
brand_id	Int	Stores the brand id of the product.
seller_id	Int	Stores the seller id who added the product.

15. Table Name: tbl_mealtype

Description: This table is used to store different types of meals along with their serving time range. It helps in classifying recipes based on meal timing such as breakfast, lunch, and dinner.

Field Name	Data Type	Description
mealtype_id	Int	Stores a unique identifier for each meal type.
mealtype_name	Varchar(50)	Stores the name of the meal type (Breakfast, Lunch, Dinner, etc.).
fromtime	Time	Stores the starting time of the meal type.
totime	Time	Stores the ending time of the meal type.

16. **Table Name: tbl_deliveryboy**
Description: This table stores the details of delivery boys responsible for delivering products to users. Each delivery boy is assigned to a particular place for efficient delivery management.

Field Name	Data Type	Description
deliveryboy_id	Int	Stores a unique identifier for each delivery boy.
name	Varchar(50)	Stores the full name of the delivery boy.
email	Varchar(100)	Stores the email address used for login and communication.
contact	Bigint	Stores the mobile number of the delivery boy.
address	Varchar(50)	Stores the residential address of the delivery boy.
photo	Varchar(200)	Stores the profile image of the delivery boy.
proof	Varchar(200)	Stores the identity proof submitted by the delivery boy.
password	Varchar(50)	Stores the login password of the delivery boy.
status	Varchar(20)	Stores the working/approval status.
doj	Date	Stores the date of joining.
place_id	Int	Stores the place id of the delivery boy.

17. **Table Name: tbl_recipe**
Description: This table stores complete information about recipes. Each recipe is linked with a meal type and food category, making it easy for users to browse and select recipes.

Field Name	Data Type	Description
recipe_id	Int	Stores a unique identifier for each recipe.
recipe_name	Varchar(100)	Stores the name of the recipe.
description	Varchar(100)	Stores the preparation details of the recipe.
image	Varchar(200)	Stores the image of the recipe.
mealtype_id	Int	Stores the meal type id related to the recipe.
foodcategory_id	Int	Stores the food category id of the recipe.

18. **Table Name: tbl_ingredients**
Description: This table stores ingredient details used in recipes. It helps in maintaining a list of ingredients required for preparing each recipe along with additional remarks or notes.

Field Name	Data Type	Description
ingredient_id	Int	Stores a unique identifier for each ingredient.
ingredient_name	Varchar(100)	Stores the name of the ingredient.
quantity	Varchar(50)	Stores the quantity or measurement of the ingredient.
remark	Varchar(100)	Stores additional notes or remarks related to the ingredient (optional).
recipe_id	Int	Stores the recipe id to which the ingredient belongs.

19. **Table Name: tbl_rating**
Description: This table is used to store ratings and reviews provided by users for recipes. It helps in evaluating recipe quality and user satisfaction.

Field Name	Data Type	Description
rating_id	Int	Stores a unique identifier for each rating.
rating_value	Int	Stores the rating value given by the user (1–5).
review	Varchar(50)	Stores the review or comments of the user.
rating_date	Date	Stores the date on which rating is given.
user_id	Int	Stores the user id who gave the rating.
recipe_id	Int	Stores the recipe id which is being rated.
product_id	Int	Foreign key referencing the product being rated.

20. **Table Name: tbl_feedback**
Description: This table stores feedback provided by users about the system and services.

Field Name	Data Type	Description
feedback_id	Int	Stores a unique identifier for each feedback.
content	Varchar(50)	Stores feedback message given by the user.
date	Date	Stores the feedback submission date.
user_id	Int	Stores the user id who submitted feedback.

21. Table Name: tbl_booking

Description: This table stores the booking/order details of users.

Field Name	Data Type	Description
booking_id	Int	Stores a unique identifier for each booking.
status	Varchar(20)	Stores the current status of the booking.
date	Date	Stores the booking date.
amount	Double	Stores the total amount paid by the user.
user_id	Int	Stores the user id who placed the booking.
deliveryboy_id	Int	Stores the assigned delivery boy id.

22. Table Name: tbl_complaint

Description: This table stores complaints raised by users regarding products or services.

Field Name	Data Type	Description
complaint_id	Int	Stores a unique identifier for each complaint.
title	Varchar(100)	Stores the title of the complaint.
content	Text	Stores the detailed complaint message.
status	Varchar(20)	Stores the complaint status (pending/resolved).
date	Date	Stores the complaint registered date.
reply	Text	Stores the admin reply to the complaint.
user_id	Int	Stores the user id who raised the complaint.